Auditing a Deployed LLM Judge for Data Matching: Is Any Weighting of Its Ingredients Better Than the One It Ships With?

2026

1 Setup

1.1 Suite and pools

- 31 datasets: 11 entity matching (EM), 6 schema matching (SM), 11 table question answering and understanding (TQA), 3 financial documents (FIN). Each adapter reads the official test or dev split and emits items with a question, a context (table, document, or record pair), a gold answer with aliases, a list gold where the answer is a set, an answer type (numeric / boolean / text / free-form) and the dataset's native metric.
- Pools: 150 items per dataset by seeded stratified sampling (strata: question type, sub-task, database, product category, match label); 100 flagged for answering (OfficeQA Pro V2: all 90; Alaska schema: 93). The remaining 50 are a seeded reserve and are not used by any step. Total answered items: 3,083 per model.
- Entity-matching pools are 50/50 match / non-match (Papadakis–Christen keeps its real 1.8 % match ratio). Schema matching is framed as one question per source column with a candidate list that always contains the gold. BIRD is graded on the executed result of the gold SQL, not on SQL text.
- Official census: every dataset's official population is counted by the same strata, so that pool numbers can be re-weighted to the official split (Section 1.5).

1.2 Collecting answers

- One prompt per item: context, question, and an answer-format line requesting `FINAL ANSWER: <answer>`.
- Models: Claude Opus 5, Claude Sonnet 5, GPT-5.5, GPT-5.4-mini, Gemini 3.8 Flash, Grok 4, DeepSeek-chat, DeepSeek-R1, gpt-oss-120b (Together), Llama 3.3 70B (Together). Gemini 3.1 Pro is collected separately at 240 items a day by a scheduled GitHub Actions workflow and enters the study automatically at 90 % suite coverage.
- Decoding: temperature 0 where the API accepts it (Gemini, Grok, DeepSeek, Llama, gpt-oss). Claude 5 reports the parameter deprecated and GPT-5.5 rejects any non-default value; those ran at provider defaults. Every cached row records which applied.
- Output budget 700 tokens. A reply cut off for length with no final-answer line is re-asked at 2,500 tokens, then at 8,000. The trigger is mechanical (finish reason plus absence of the final line), never correctness. Originals are archived with a reason; replacement rows carry `attempt` = 2 or 3. Replies still unterminated at 8,000 tokens are scored as given (effectively failures).

- Every answer is cached with tokens, latency, cost, finish reason and decoding; re-runs skip cached items. Cost: USD 161 for ten models (Claude Opus 5: USD 99).

1.3 Scoring an answer

- Extraction: the `FINAL ANSWER:` line; else a boxed expression; else the last short line of the reply.
- Normalisation: Unicode NFKC, lower-case, canonical numbers ("1,577.00" → "1577", "(12.6)" → "−12.6"), currency symbols and punctuation removed, articles dropped, whitespace collapsed.
- Units and scale: every number is parsed with sign, percent marker (%, "percent", "pp") and scale word (thousand / million / billion / trillion and abbreviations). A parsed number exposes its equivalent readings (raw, scaled, ÷100 if marked percent, ×100 if an unmarked fraction); the relative error between prediction and gold is the minimum over all pairs of readings. Real-unit conversions (km ↔ miles) are not attempted.
- Aliases: each item carries the primary gold plus acceptable alternatives supplied by the adapter (e.g. *yes* ≡ *match* ≡ 1; several legitimate target columns in schema matching; comma- and pipe-joined forms of list golds). Every component is computed against each alias and the best-scoring alias is kept. Booleans are canonicalised before comparison.
- Components per answer: token F1, numeric decay $e^{-2.5 \cdot \text{rel. err}}$ (error clipped at 2; falls back to exact match for non-numeric golds), token precision, token recall; also exact match, numeric tolerance (1 %), ROUGE-L, and a hedge flag (several distinct numeric candidates, "A or B", or an enumeration of short candidates).
- Native metric per dataset (baseline column): numeric tolerance for numeric golds, exact match for text and boolean golds, ROUGE-L for free-form (FeTaQA).
- Composite under weighting w on the simplex: $R_w = \sum_k w_k m_k$, $k \in \{\text{F1, decay, precision, recall}\}$. v1 is $w = (0.35, 0.35, 0.15, 0.15)$.
- Dataset score of a model: mean R_w over the items that every model answered. A model is in the study only if it covered ≥ 90 % of the suite; a dataset is used only with ≥ 20 answered items.

1.4 Evaluating a weighting

1. **Agreement $J(w)$.** Rank the ten models inside each dataset under w; pool the 31 rankings by Borda; J is the mean Kendall τ_b between each dataset's ranking and the pooled one. J measures consistency between datasets, not correctness. Computed for all 1,771 weightings of the step-0.05 simplex lattice (v1, the four corners and the uniform mix are lattice points), and also under Kemeny, Copeland and Bradley–Terry pooling.
2. **Correctness $\rho(w)$.** 31,380 synthetic answers of known utility generated from the golds of the answered items by typed operators (identity, alias, unit variant, typo, rounding, magnitude error, sign flip, wrong value, partial list, superset, hedge, verbose, no answer, . . .). Reported: Spearman ρ between R_w and utility, AUC correct vs wrong, mean score of wrong answers, hedge gap; the (J, ρ) Pareto front and its knee.
3. **Honest selection.** Nested item bootstrap (B = 1,000): choose the best w on in-bag items, score it out of bag. Reported: the optimism gap, the paired out-of-bag difference to v1 with a 95 % interval, and the plateau (weightings not significantly below the best, paired bootstrap $p > .05$) plus a practical plateau (within 0.02 of the best).
4. **Transfer.** Leave-one-dataset-out and leave-one-family-out: re-fit w on the rest, measure τ of the held-out dataset's ranking to the pooled rest; per-family optima; sensitivity to one lattice step (0.05).
5. **Designs.** Uniform-lattice, Dirichlet-snapped and Latin-hypercube samples of 100 weightings,

20 seeds each, compared with the exhaustive optimum (regret, rank of the found best, centred L_2 discrepancy, minimum pairwise L_1).

1.5 From scores to a leaderboard

- Rules reported: exact match only, v1, the five best by J, the best by ρ, the (J, ρ) Pareto knee.
- Pooling: Bradley–Terry strengths from item-level pairwise wins (MLE, ridge 0.01, standard errors from the observed information) as the headline; Borda, mean-z, Kemeny (with cost), Copeland, reciprocal-rank fusion and family-balanced z as checks.
- Uncertainty: 1,000 item bootstraps of the Borda and Bradley–Terry rankings (rank intervals, probability of first place), saved in full.
- Significance: Friedman with the Iman–Davenport correction across datasets, Nemenyi critical difference, Holm-corrected pairwise Wilcoxon; paired permutation and Wilcoxon per dataset and model pair (Holm within dataset); Kendall's W among datasets; majority-class baselines for boolean sets.
- Official-split estimates: the same answers re-weighted by official stratum share / sample share (post-stratification) with a weighted item bootstrap; positive-class precision, recall and F1 at the official match ratio for the matching sets.
- Diagnostics: pairwise τ between dataset rankings; floor, ceiling and self-stability flags per dataset.
- Reporting: one script regenerates the report from the cached answers in minutes, so text and numbers cannot disagree.

2 Results

All numbers are from the final run of 2026-09-27 (31 datasets × 10 models, 3,083 answered items per model). Table 1 gives the audit in one place; the subsections that follow give one figure or table per row.

2.1 The objective is flat

Figure 1 shows J for all 1,771 weightings. The whole lattice spans 0.39–0.42. The best weighting (decay 0.70, recall 0.30) reaches 0.421 and v1 reaches 0.404; 1,482 weightings are nominally better than v1, but the entire spread is smaller than the bootstrap noise of a single weighting. No single weight has a slope worth acting on (Figure 2).

2.2 The best weighting does not survive honest selection

Figure 3 reports the nested item bootstrap. The weighting chosen on in-bag items scores 0.380 in bag but 0.331 out of bag (optimism gap 0.049), below v1's out-of-bag 0.339. The paired out-of-bag difference to v1 is -0.008 with a 95 % interval of $[-0.039, +0.013]$ ($p = 0.72$); the selected weighting is ahead of v1 in 28 % of resamples, and 608 distinct weightings were selected across the 1,000 resamples. The plateau of weightings not significantly below the best contains 1,765 of 1,771 points and v1 is inside it ($p = 0.66$).

2.3 The optimum does not transfer, and neither does the rank

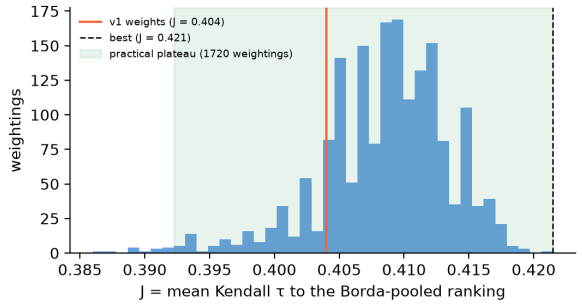
Leave-one-dataset-out: weights re-fitted without the held-out dataset predict its ranking with mean $\tau = +0.34$, the full-data best with $+0.39$, v1 with $+0.37$ (Table 3 gives the family-level version).

Table 1: Audit summary. J = agreement objective (mean Kendall τ_b of each dataset's model ranking to the Borda pool); ρ = Spearman correlation of the composite with known utility on 31,380 synthetic anchors.

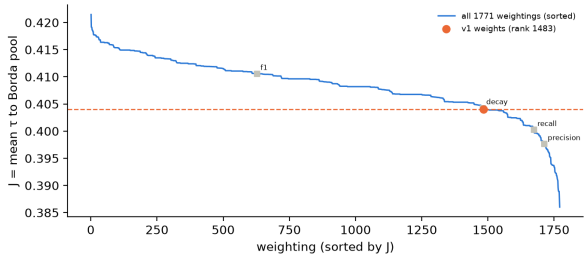
Exhaustive grid	1,771 weightings (step 0.05); best decay .70 / recall .30, $J = 0.421$; v1 $J = 0.404$ (rank 1,483 of 1,771)
J over the lattice	median 0.409, 90th percentile 0.415, 99th percentile 0.418: a flat landscape
Plateau (paired bootstrap $p > .05$ vs best)	1,765 of 1,771 (v1 inside, $p = 0.66$); practical plateau within 0.02: 1,720
Nested selection ($B = 1{,}000$)	in-bag J of the selected weighting 0.380, out-of-bag 0.331 (optimism gap 0.049); v1 out-of-bag 0.339
Selected vs v1, out of bag	-0.008, 95 % CI $[-0.039, +0.013]$, $p = 0.72$; 608 distinct weightings selected across resamples
Sampled designs (100 points, 20 seeds)	Dirichlet / LHS / uniform all reach 0.418 ± 0.001, regret ≤ 0.005: sampling would have sufficed
Objective under other pooling rules	Kemeny, Copeland and Bradley–Terry each pick a different "best"; v1's rank swings from 393 to 1,506: rank is noise
Transfer (leave-one-dataset-out)	held-out τ: LODO weights +0.34, full best +0.39, v1 +0.37
Leave-one-family-out	v1 within 0.02 of the held-out-fitted weights on EM, SM, TQA; FIN favours precision-heavy weights (+0.51 vs +0.43)
Anchors (31,380 synthetic answers)	v1 $\rho = 0.878$, best-ρ weighting 0.883, best-J weighting 0.845, exact match 0.828; AUC correct-vs-wrong 0.994 for every rule
J vs ρ across the lattice	-0.34 on this cache; +0.38 on the previous cache: unstable because J is flat
Sensitivity	one 0.05 step moves J by at most 0.007 around v1 and 0.005 around the best
Leaderboard	Friedman $p < 0.001$ under every rule; Kendall's W among datasets 0.26–0.28
Significant pairs (Holm)	Opus, Sonnet, GPT-5.5, Gemini Flash, DeepSeek-chat, DeepSeek-R1, GPT-5.4-mini, Grok 4 all > Llama 3.3 70B; Opus, Gemini Flash, GPT-5.5, Grok 4 > gpt-oss-120b
First place	Bradley–Terry: Gemini 3.8 Flash under v1 and the calibrated rules ($P(\text{first}) = 0.99$), Opus under exact match; Borda: GPT-5.5 under v1, Opus under exact match and the top-5 rules
Official-split comparability	accuracy moves by ≤ 0.09 after post-stratification; positive-class F1 on the balanced pools overstates official-ratio F1 by up to 0.19 (Amazon–Google)
Datasets at ceiling	Valentine, WDC LSPC, WDC Products (means 0.98): flagged; still inside the objective
Cost	USD 161 total (Opus 99)

Table 2: Sampled designs at a budget of 100 weightings (20 seeds each) against the exhaustive optimum.

Design	best J (mean $\pm$ sd)	regret mean / max	rank of found best (median / worst)	min pairwise L_1	centred L_2 discrepancy	share with a zero weight
Dirichlet-snapped	0.418 ± 0.001	0.004 / 0.005	14 / 49	0.10	0.538	0.27
Latin hypercube	0.418 ± 0.001	0.003 / 0.005	10 / 49	0.10	0.531	0.28
Uniform lattice	0.418 ± 0.001	0.003 / 0.005	6 / 42	0.10	0.526	0.45



(a) Distribution of J over the lattice with v1, the best and the plateau band.



(b) Sorted J with v1 and the four single-ingredient corners.

Figure 1: The agreement landscape over all 1,771 step-0.05 weightings.

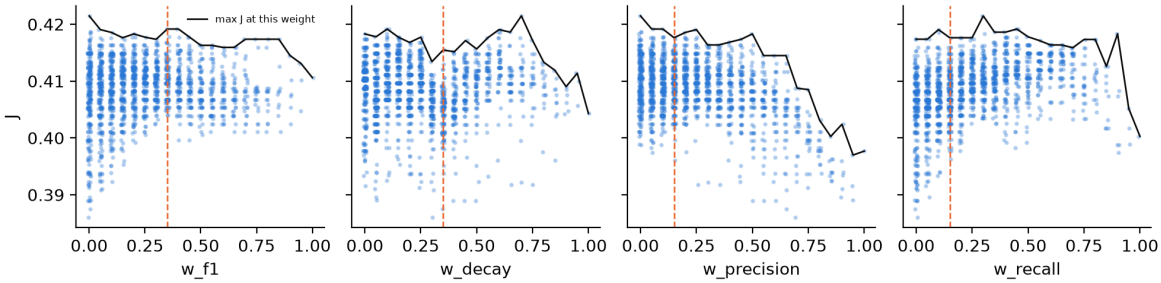
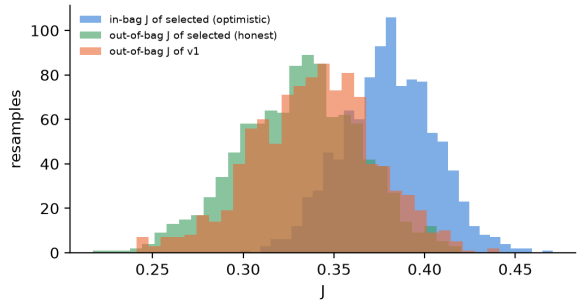
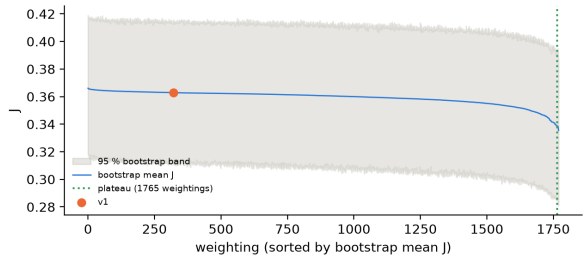


Figure 2: J against each of the four weights. Corners: F1 only 0.411, decay only 0.404, precision only 0.398, recall only 0.400; uniform 0.410.



(a) In-bag vs out-of-bag J of the selected weighting against v1.



(b) Bootstrap band over sorted weightings; the plateau boundary.

Figure 3: Honest selection: the optimism gap and the plateau.

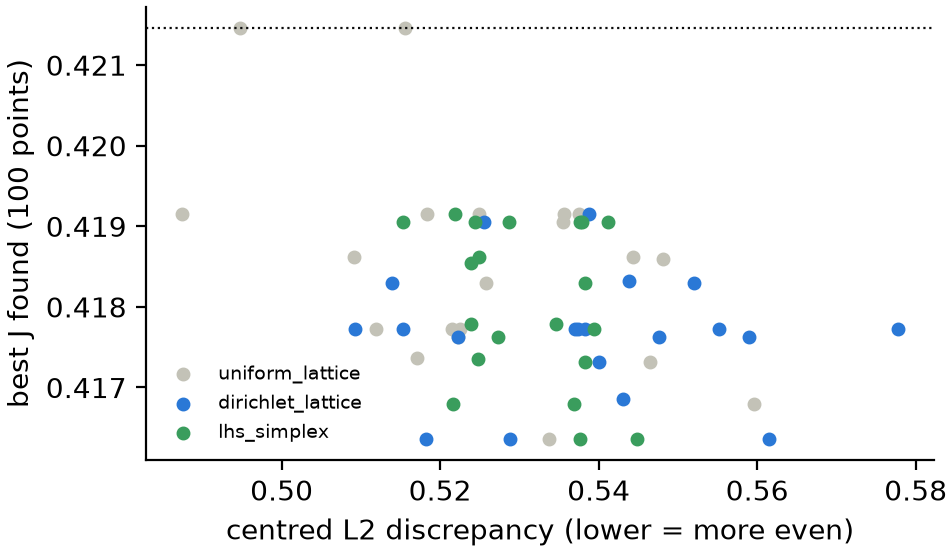


Figure 4: Any 100-point design finds the plateau; regret against the exhaustive optimum is at most 0.005.

Per-family optima pull in different directions (Figure 5): decay-dominated for EM and SM, recall-dominated for TQA, precision-heavy for FIN, yet none is far from the plateau. Changing only the pooling rule inside the objective moves v1's rank from 393 (Bradley–Terry) to 1,506 (Kemeny) and changes which weighting is "best", so rank inside a flat objective carries no information.

Table 3: Leave-one-family-out. τ = agreement of the held-out family's pooled ranking with the pooled ranking of the remaining families.

Held-out family	LOFO weights (F1 / decay / prec / rec)	τ: LOFO weights	full best	v1
EM (11 sets)	0.20 / 0.00 / 0.50 / 0.30	+0.32	+0.29	+0.30
SM (6 sets)	0.00 / 0.10 / 0.00 / 0.90	+0.35	+0.37	+0.25
TQA (11 sets)	0.45 / 0.00 / 0.40 / 0.15	+0.38	+0.40	+0.38
FIN (3 sets)	0.00 / 0.95 / 0.05 / 0.00	+0.43	+0.51	+0.51

2.4 Agreement and correctness do not pull the same way

On 31,380 synthetic answers of known utility, v1 scores $\rho = 0.878$; the best weighting by ρ reaches 0.883 and the best by J only 0.845; exact match 0.828. Every rule separates correct from wrong answers with AUC 0.994. Across the lattice J and ρ correlate -0.34 on this cache and $+0.38$ on the previous one, which is what a flat J predicts. The Pareto knee (F1 0.05, decay 0.60, precision 0.25, recall 0.10) is within 0.002 of the best J and within 0.006 of the best ρ (Table 4, Figure 6).

2.5 Leaderboard

Under every rule the Friedman test rejects equal models ($p < 0.001$) while datasets agree only moderately (Kendall's W 0.26–0.28). The ends of the ranking are stable: Llama 3.3 70B is last and gpt-oss-120b ninth under every rule and every pooling method, with Holm-corrected significance against the top group. The order inside the top three depends on the rule and the pooling method

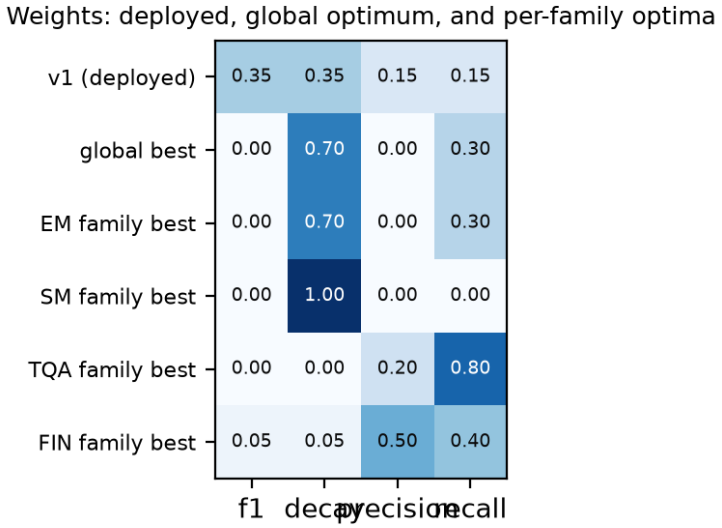


Figure 5: Deployed weights (v1), the global optimum and the per-family optima.

Table 4: Correctness on known-quality anchors (9,881 correct, 7,430 wrong, remainder partial). ρ_d = mean per-dataset ρ; wrong = mean score given to wrong answers.

Rule	F1 / decay / prec / rec	J	ρ	ρ_d	AUC	wrong	hedge gap
v1	0.35 / 0.35 / 0.15 / 0.15	0.404	0.878	0.868	0.994	0.034	−0.51
best J	0.00 / 0.70 / 0.00 / 0.30	0.421	0.845	0.840	0.994	0.037	−0.49
best ρ	0.00 / 0.35 / 0.60 / 0.05	0.398	0.883	0.871	0.994	0.034	−0.57
Pareto knee	0.05 / 0.60 / 0.25 / 0.10	0.419	0.877	0.866	0.994	0.036	−0.55
exact match	–	–	0.828	0.830	0.989	0.011	−0.81

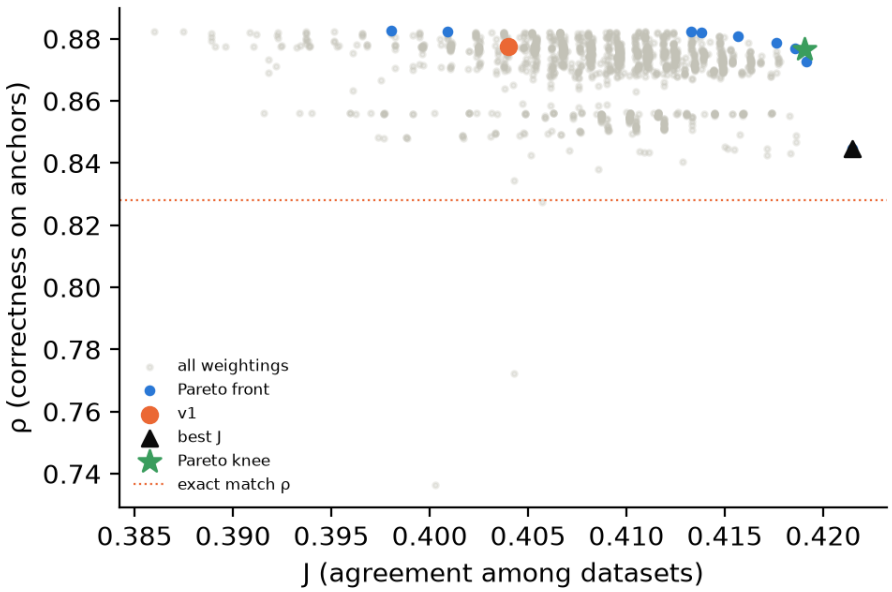


Figure 6: Agreement J against anchor correctness ρ for every weighting; v1, the best-J, best-ρ and knee points marked.

(Table 5, Table 6, Figure 7): Bradley–Terry puts Gemini 3.8 Flash first under v1 and all calibrated rules, Borda puts GPT-5.5 first under v1 and Opus first under the top-5 rules, and exact match puts Opus first under both.

Table 5: Leaderboard under the deployed rule (v1 weights). Bradley–Terry strength ± SE relative to Opus; Borda rank with 95 % bootstrap interval; mean composite score over the suite.

Rank	Model	BT strength	Borda rank [95 %]	mean score
1	Gemini 3.8 Flash	$+0.12 \pm 0.02$	3 [1, 3]	0.812
2	Claude Opus 5	$+0.10 \pm 0.00$	2 [1, 3]	0.811
3	GPT-5.5	$+0.05 \pm 0.02$	1 [1, 3]	0.808
4	Claude Sonnet 5	$+0.03 \pm 0.02$	4 [4, 6]	0.795
5	Grok 4	$+0.03 \pm 0.02$	5 [4, 6]	0.797
6	GPT-5.4-mini	-0.01 ± 0.02	8 [5, 8]	0.789
7	DeepSeek-chat	-0.02 ± 0.02	7 [5, 8]	0.783
8	DeepSeek-R1	-0.03 ± 0.02	6 [5, 8]	0.788
9	gpt-oss-120b	-0.10 ± 0.02	9 [9, 10]	0.767
10	Llama 3.3 70B	-0.17 ± 0.02	10 [9, 10]	0.739

Table 6: Bradley–Terry rank (Borda rank in brackets) under each reported rule. top1–top5 = the five best weightings by J; best-ρ = best by anchor correctness; knee = (J, ρ) Pareto knee.

Rule	Opus 5	Sonnet 5	DS-chat	DS-R1	Flash 3.8	5.4-mini	GPT-5.5	gpt-oss	Grok 4	Llama
exact match	1 (1)	5 (4)	8 (7)	7 (6)	3 (3)	6 (8)	2 (2)	9 (9)	4 (5)	10 (10)
v1	2 (2)	4 (4)	7 (7)	8 (6)	1 (3)	6 (8)	3 (1)	9 (9)	5 (5)	10 (10)
top1	2 (1)	4 (4)	7 (8)	6 (6)	1 (3)	8 (7)	3 (2)	9 (9)	5 (5)	10 (10)
top2	2 (1)	4 (4)	6 (7)	8 (6)	1 (3)	7 (8)	3 (2)	9 (9)	5 (5)	10 (10)
top3	2 (1)	4 (4)	6 (7)	8 (6)	1 (3)	7 (8)	3 (2)	9 (9)	5 (5)	10 (10)
top4	2 (1)	4 (4)	7 (8)	8 (6)	1 (3)	6 (7)	3 (2)	9 (9)	5 (5)	10 (10)
top5	2 (1)	4 (4)	7 (8)	6 (6)	1 (3)	8 (7)	3 (2)	9 (9)	5 (5)	10 (10)
best-ρ	2 (2)	4 (4)	8 (8)	7 (6)	1 (3)	5 (7)	3 (1)	9 (9)	6 (5)	10 (10)
knee	2 (1)	4 (4)	7 (8)	8 (6)	1 (3)	6 (7)	3 (2)	9 (9)	5 (5)	10 (10)

Kendall τ between the BT rankings of the rules: 0.78–1.00 (exact match vs the rest 0.78–0.82).

2.6 Comparability with published numbers

Post-stratification to the official split changes dataset accuracy by at most 0.09 (OpenSanctions pairs 0.82 → 0.73; TAT-QA 0.80 → 0.76; MMTU 0.87 → 0.83). For the matching sets, positive-class F1 computed on the balanced 50/50 pool overstates F1 at the official match ratio wherever the official split is sparse (Table 7, Figure 9).

2.7 Dataset diagnostics

Three datasets sit at ceiling (mean composite 0.98 with a spread of 0.02–0.04 across models): Valentine, WDC LSPC and WDC Products. They are flagged and remain inside the objective in this run. Two sit near floor for every model (OfficeQA closed-book 0.13, OfficeQA Pro V2 0.14) and one is a known hard set (BIRD 0.19); none is excluded. Self-stability τ (rank agreement between split halves of a dataset) ranges from 0.58 (HiTab) to 0.93 (FeTaQA).

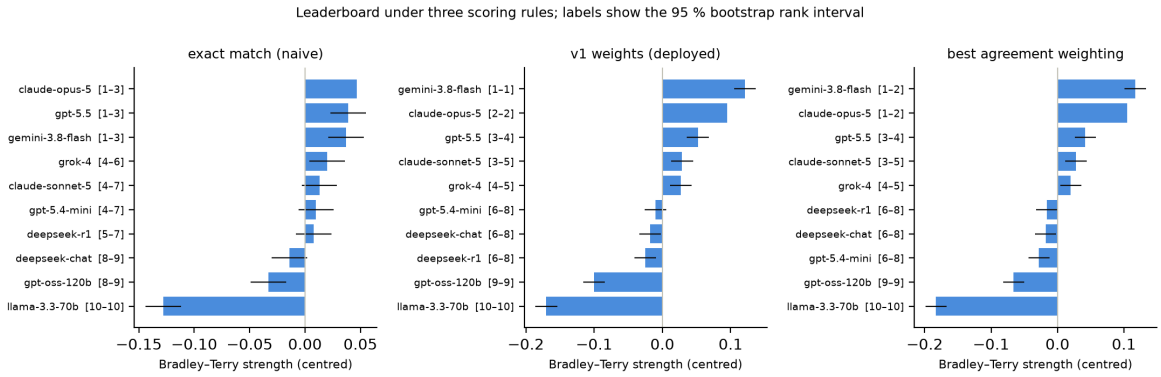


Figure 7: Bradley–Terry strengths ± SE with bootstrap rank intervals under exact match, v1 and the best-J weighting. The ends are stable; the top order is rule-dependent.

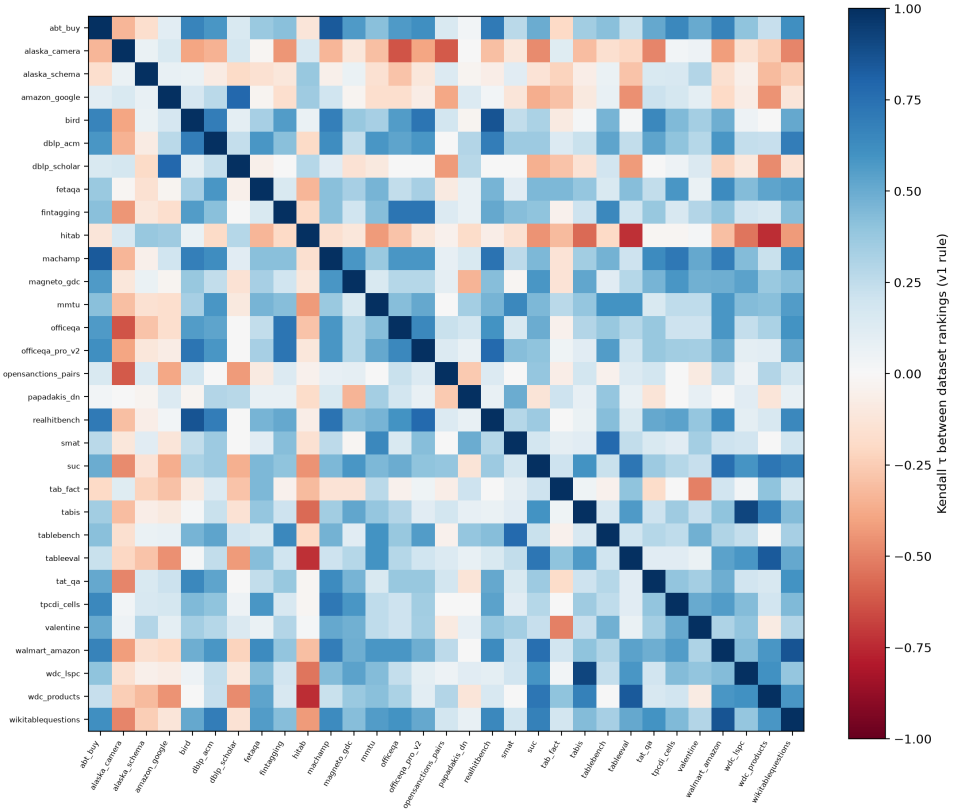


Figure 8: Kendall τ between the model rankings of each pair of datasets under v1. Datasets agree only moderately ($W \approx 0.27$).

Table 7: Matching sets under v1, averaged over the ten models: suite accuracy, post-stratified official accuracy, and positive-class F1 on the balanced pool vs at the official match ratio.

Dataset	official match share	suite acc.	official acc.	F1 (pool)	F1 (official ratio)
Abt–Buy	0.108	0.970	0.986	0.968	0.936
Amazon–Google	0.102	0.936	0.932	0.929	0.739
Walmart–Amazon	0.094	0.958	0.975	0.947	0.875
DBLP–ACM	0.180	0.986	0.991	0.985	0.976
DBLP–Scholar	0.186	0.949	0.966	0.948	0.910
WDC Products 2024	0.280	0.978	0.985	0.979	0.974
WDC LSPC 2017	0.264	0.982	0.983	0.983	0.971
Machamp	0.175	0.887	0.890	0.892	0.733
Papadakis–Christen Dn	0.018	0.941	0.988	0.833	0.683
OpenSanctions pairs	0.769	0.821	0.727	0.773	0.777
Alaska Camera	n/c	0.958	–	0.955	–

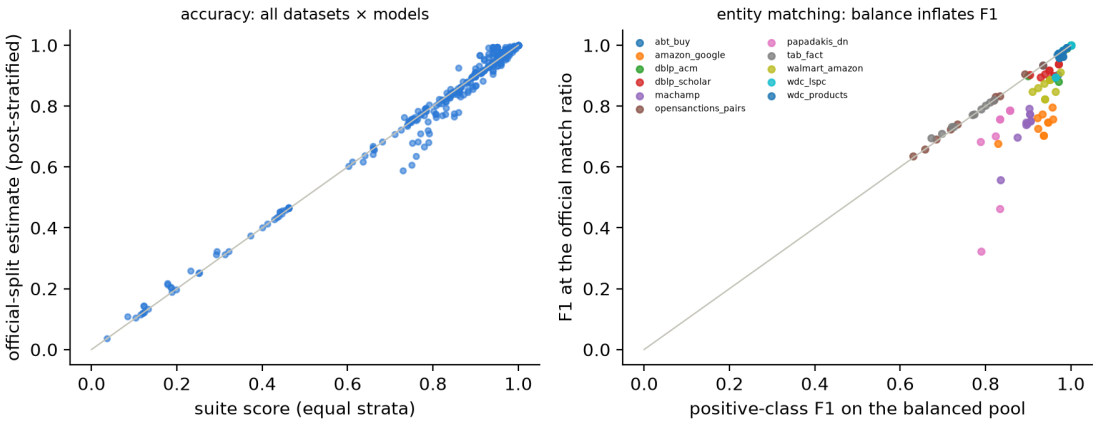


Figure 9: Suite score vs post-stratified official estimate per dataset, and matching F1 on the balanced pool vs at the official match ratio.